

Parallel lines and transversals

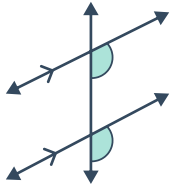


1

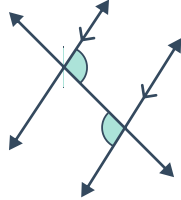
- a How are parallel lines or segments marked on a diagram?
- b Draw an example of correctly marked parallel lines.

2 Identify the types of angles formed by the transversal in the following diagrams:

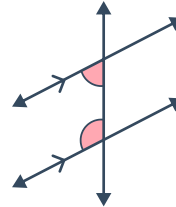
a



b



c



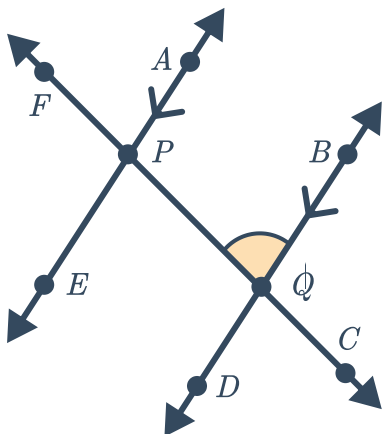
3 State whether the following types of angles are equal, complementary, or supplementary:

- a Alternate angles
- b Cointerior angles
- c Corresponding angles

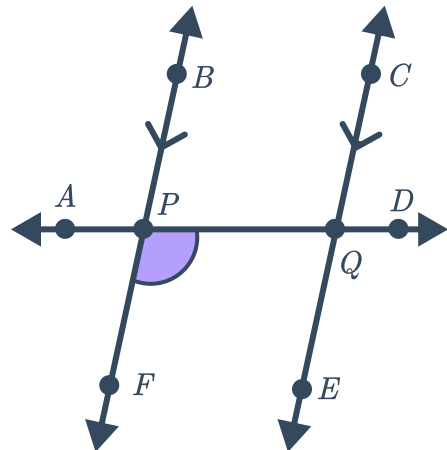
4 For the marked angle in each the following diagrams, name the angle which is:

- i Alternate
- ii Cointerior
- iii Corresponding

a



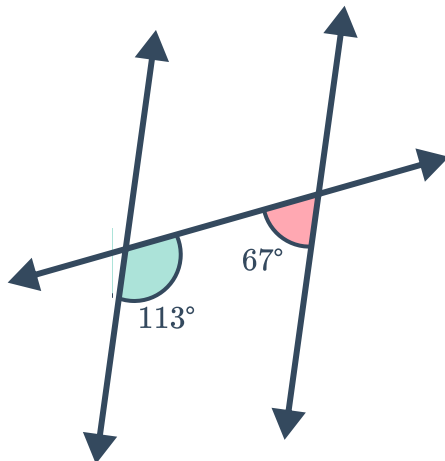
b



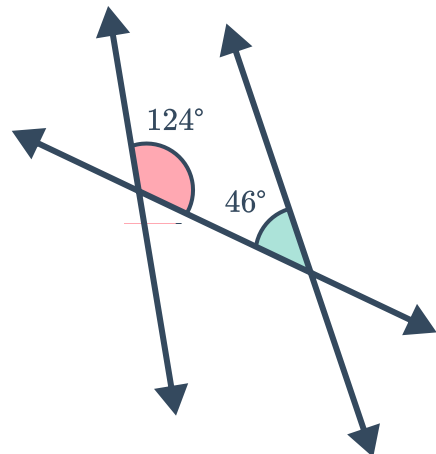
- 5 For each of the following diagrams, state whether the diagram contains parallel lines. Explain your answer.



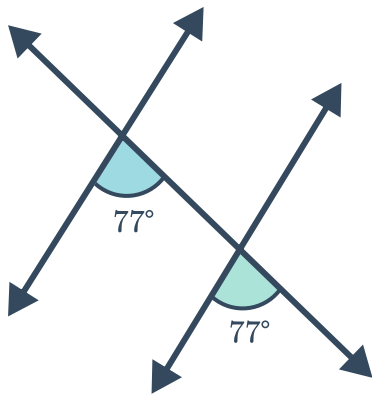
a



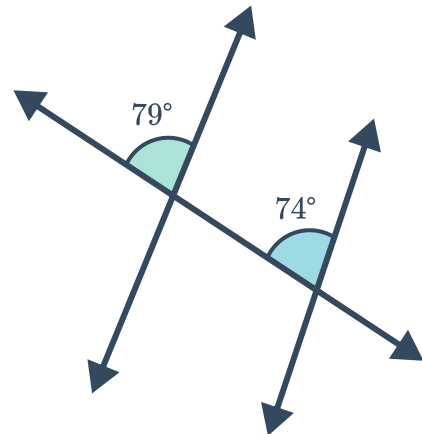
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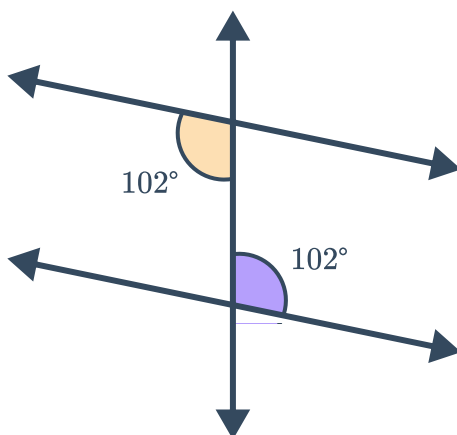
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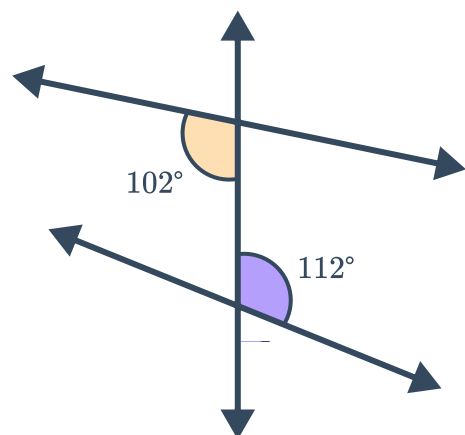
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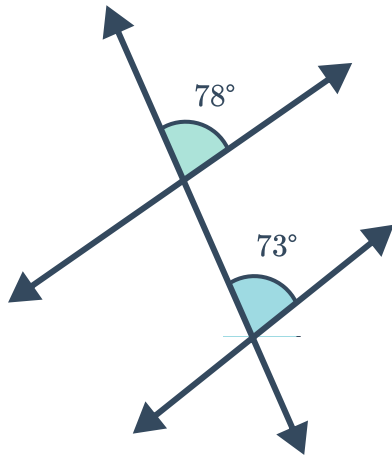
e



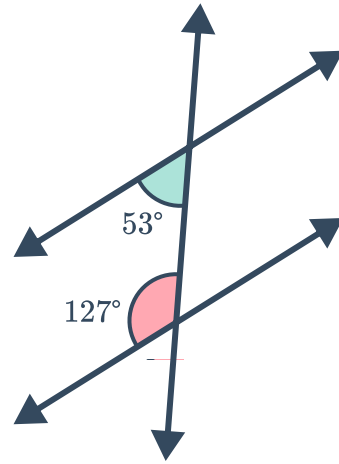
f



g

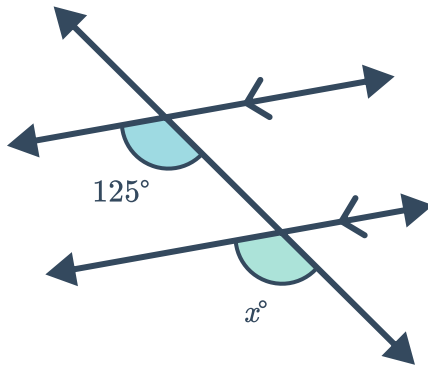


h

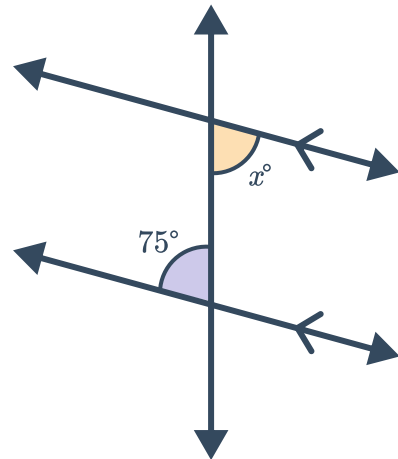


6 For each of the following diagrams, find the value of x :

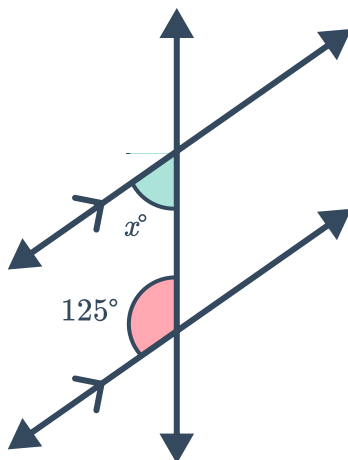
a



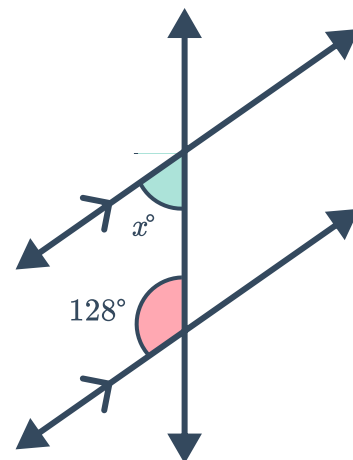
b



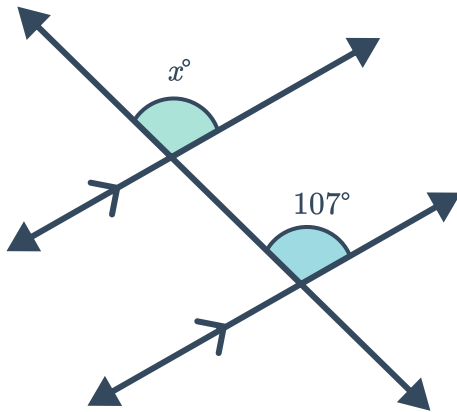
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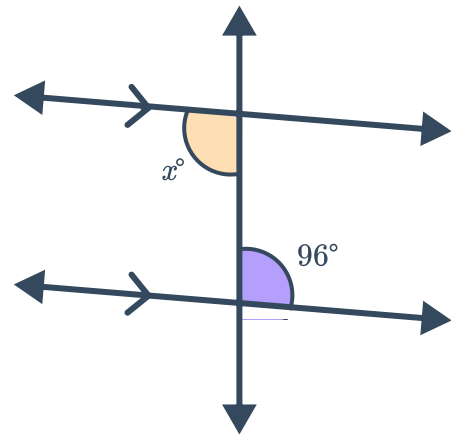
d



e

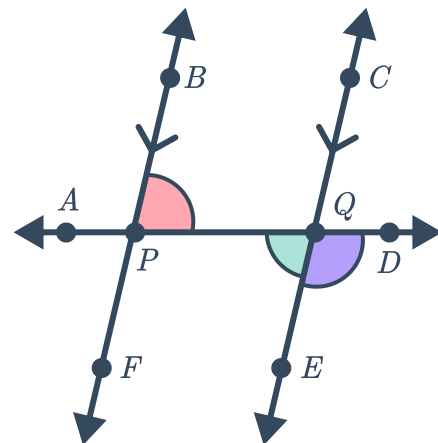


f



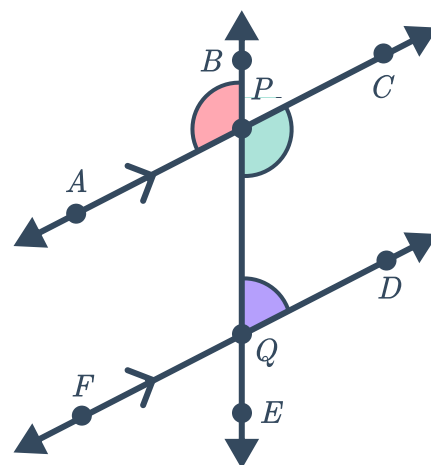
7 Consider the following diagram:

- a How are $\angle BPD$ and $\angle AQE$ related?
- b How are $\angle AQE$ and $\angle DQE$ related?
- c If $\angle BPD = 84^\circ$, find the value of $\angle DQE$.



8 Consider the following diagram:

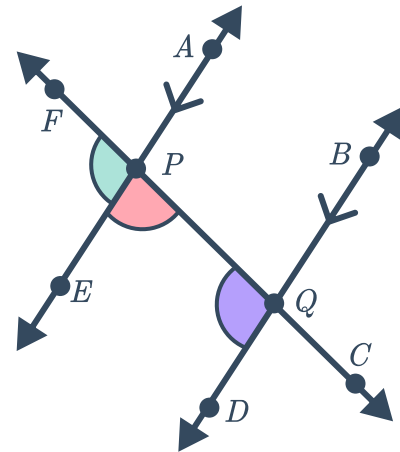
- a How are $\angle APB$ and $\angle CPE$ related?
- b How are $\angle CPE$ and $\angle BQD$ related?
- c If $\angle APB = 107^\circ$, find the value of $\angle BQD$.



9 Consider the following diagram:



- a How are $\angle DQF$ and $\angle CPE$ related?
- b How are $\angle CPE$ and $\angle EPF$ related?
- c If $\angle DQF = 97^\circ$, find the value of $\angle EPF$.



10 Consider the following diagram:

- a Find the value of x .
- b Find the value of y .

